

Titanic Dataset - EDA Findings Report

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1. Introduction

This report summarizes the exploratory data analysis performed on the Titanic dataset to understand factors influencing passenger survival.

2. Data Overview

- Total passengers: 891
- Features analyzed: 12 (including survival status, class, age, gender)
- Missing values handled:
 - Age: 177 missing (filled with median)
 - Cabin: 687 missing (column dropped)
 - Embarked: 2 missing (filled with mode)

3. Key Findings

3.1 Demographic Insights

! [Passenger Distribution by Class]

- Class distribution:
 - 1st class: 24%
 - 2nd class: 21%
 - 3rd class: 55%
- Gender ratio: 65% male, 35% female
- Age distribution: Majority aged 20-40 years

3.2 Survival Analysis

- **Overall survival rate: 38.4%**
- **Survival by class:**
 - 1st class: 63% survived
 - 2nd class: 47% survived
 - 3rd class: 24% survived
- **Survival by gender:**
 - Female: 74% survival rate
 - Male: 19% survival rate
- **Children (<10 years): 59% survival rate**

4. Visualizations Summary

1. Histograms showed:

- Normal age distribution
- Right skew for fare prices

2. Boxplots revealed:

- Higher fares among survivors
- Younger age among survivors

3. Heatmap correlations:

- Pclass vs Fare: -0.55
- Fare vs Survival: 0.26
- Pclass vs Survival: -0.34

5. Conclusion

The analysis confirms that passenger class, gender, and fare paid were significant survival factors. Women, children, and higher-class passengers had substantially better survival rates.